

NEW RECORDS OF EOCENE-OLIGOCENE MAMMALS FROM THE GRAVELLY RANGE, SOUTHWEST MONTANA

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ABSTRACT

Based on recently recovered specimens from Eocene-Oligocene strata at Black Butte in the Gravelly Range, new records of *Ischyromys* sp. cf. *I. veterior*, *Cylindrodon* sp. cf. *C. fontis*, *Eumys cricetodontoides*, *Hypisodus minimus*, and *Leptomeryx speciosus* are known from two new mammalian assemblages, Black Butte Middle I and Black Butte Middle II. These assemblages are in strata above the late Uintan Black Butte Low assemblage and below the Black Butte High assemblage now considered Orellan because of the addition of *Eumys cricetodontoides*. Black Butte Middle I strata yielded *Ischyromys* sp. cf. *I. veterior*, *Cylindrodon* sp. cf. *C. fontis*, and *Leptomeryx speciosus*, taxa best known from middle Chadronian rocks in southwest Montana. Black Butte Middle II strata are about 45 m above the Black Butte Middle I mammalian assemblage and yielded *Leptomeryx speciosus* and *Hypisodus minimus*, the second known occurrence of *H. minimus* in Montana other than late Chadronian rocks in the Three Forks area. The presence of *Leptomeryx speciosus* indicates that Black Butte Middle II is middle-late Chadronian. Documentation of four late Uintan-Orellan mammalian assemblages at Black Butte combined with the ages of assemblages previously identified at Lion Mountain (Whitneyan), Teepee Mountain (late Duchesnean-early Chadronian), and the *Rapamys* (late Uintan) indicate that Tertiary sedimentation in the Gravelly Range area apparently occurred throughout the middle Eocene-early Oligocene (late Uintan to Whitneyan) without a major hiatus.

INTRODUCTION

Tertiary strata yield significant but mostly sparse assemblages of mammals at high elevation areas (approximately 3,000 m) in the south-central part of the Gravelly Range, southwest Montana (Figure 1). These rocks are part of the informal Sequence 2 of Hanneman and Wideman (1991, 2006) and a lithostratigraphic proxy, the Renova Formation of Kuenzi and Fields (1971). In the late 1940s John Mann collected mammalian fossils at Black Butte while studying the geology of the Gravelly Range for his PhD dissertation from Princeton University where Professor Glenn Jepsen's analysis indicated that Black Butte strata were probably early Oligocene in age (Mann 1960). Other collections of vertebrates from the Gravelly Range were made by Arnie Lewis from the Museum of Comparative Zoology (MCZ) in the 1950s, Malcolm McKenna from the American Museum of Natural History (AMNH) in the 1980s, and Eric Luikart in the 1990s as part

of his master's thesis at Montana State University (Gutman et al. 1989; Luikart 1997). Overall, significant collections of fossil mammals were recovered from four areas of outcrop (Lion Mountain, Black Butte, Teepee Mountain, *Rapamys* Site), but the only specimens described were those representing a new species of rodent, *Rapamys atramontis* (Wahlert et al. 2006). Based on the AMNH and MCZ specimens and those collected by the Raymond M. Alf Museum of Paleontology (RAM) starting in 2016, Lofgren et al. (2020) described over twenty mammalian taxa which indicated Tertiary strata in the Gravelly Range were late Uintan to Whitneyan in age (middle Eocene to early Oligocene). The iguanid squamate *Aciprion formosum* was also described from Whitneyan strata at Lion Mountain (Lofgren et al. 2023).

Black Butte mammalian specimens collected by RAM crews since 2020

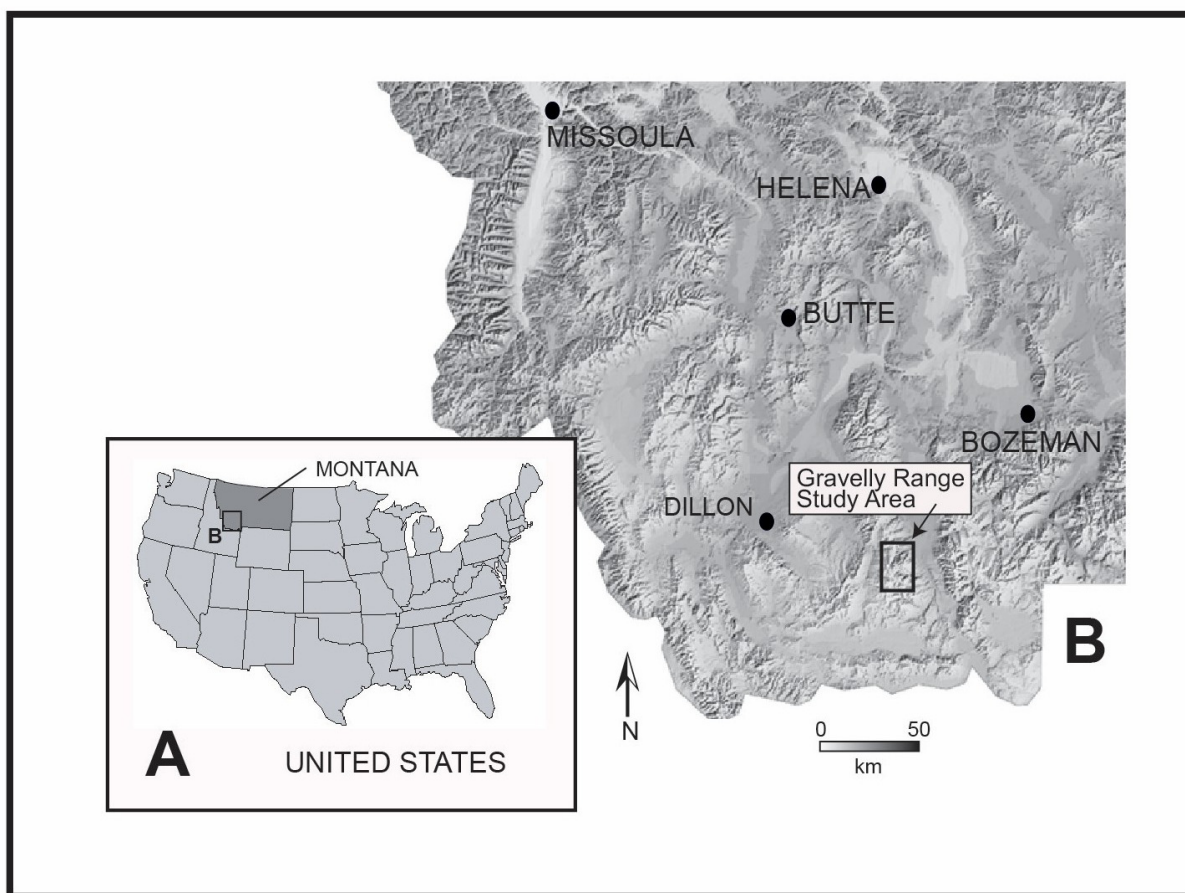


FIGURE 1. Location of middle Eocene to early Oligocene strata (Uintan to Whitneyan, North American Land Mammal Ages) in the south-central Gravelly Range of southwest Montana.

indicate that small but definitive middle and middle to late Chadronian assemblages are present (Black Butte Middle I and Black Butte Middle II) between the Black Butte Low and Black Butte High assemblages (Figure 2). The identification of four superposed mammalian assemblages at Black Butte provides new details concerning the timing of sediment deposition in the Gravelly Range area of Montana during the middle Eocene to early Oligocene.

MATERIALS AND METHODS

One Black Butte mammal specimen found by J. Mann was identified by G.

Jepsen at Princeton University as *Leptomeryx evansi* (Mann 1960). The Princeton University vertebrate paleontology collections were later acquired by the Yale Peabody Museum (YPM) and in 2025 D. Lofgren visited the YPM and the uncatalogued specimen was located. It was assigned YPM-VPPU 30630 and included in this study. The recovery of a dentary of *Hypsiodus* from the Gravelly Range also inspired inclusion of three CM specimens of *Hypsiodus* (CM 78277, CM 78278, CM 78279) reported by Tabrum (1998) from the Three Forks area. For each specimen described, a specific RAM or CM locality number is provided in Systematic

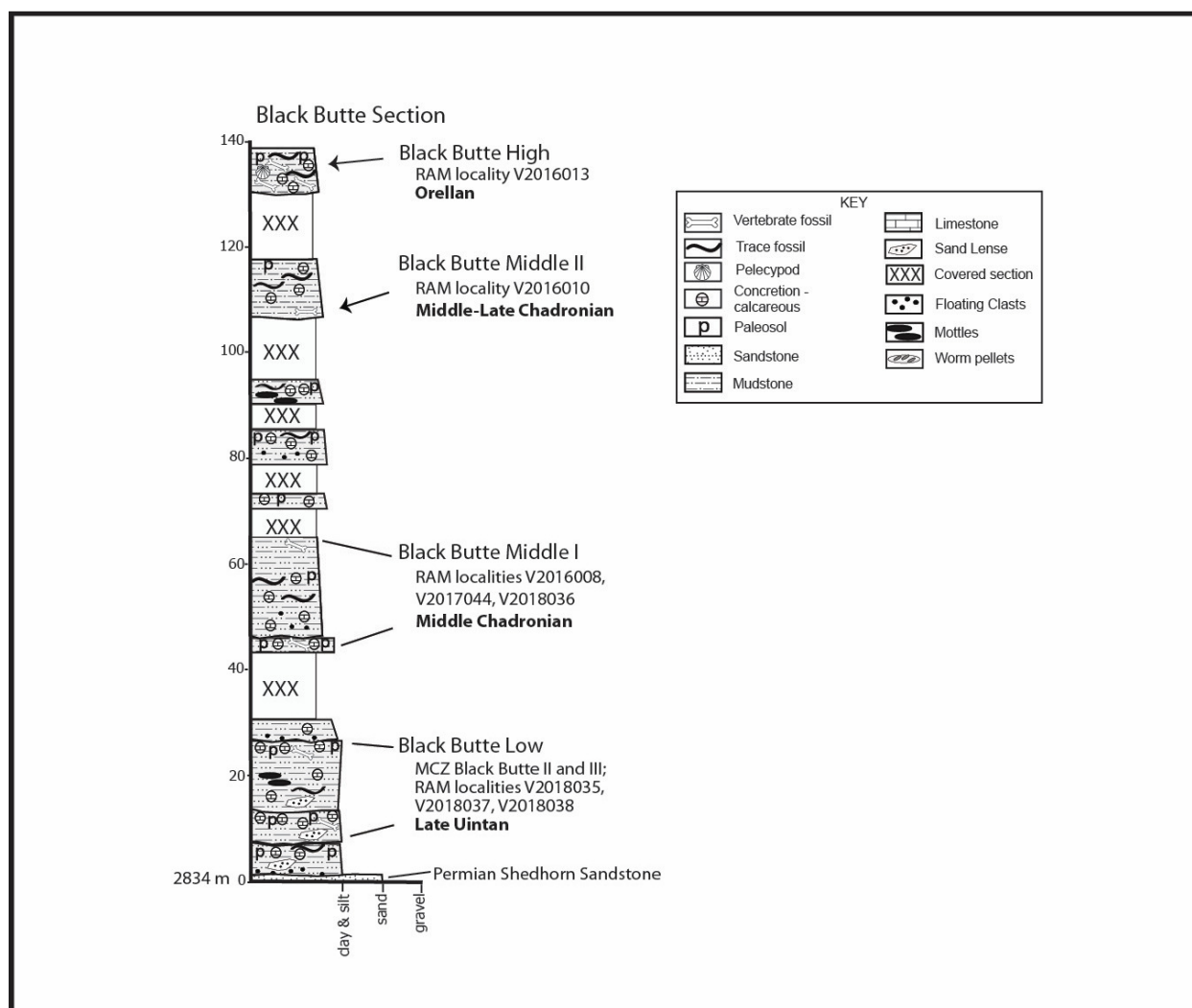


FIGURE 2. Late Uintan to Orellan mammalian assemblage locations within a measured stratigraphic section on the southeast flank of Black Butte in the Gravelly Range study area.

Paleontology. The names of Black Butte mammalian assemblages (with RAM localities) are listed below in ascending stratigraphic order (Figure 2).

--Black Butte Low (V2018035, V2018037, V2018038), late Uintan

--Black Butte Middle I (V2016008, V2017044, V2018036) middle Chadronian

--Black Butte Middle II (V2016010) middle-late Chadronian

--Black Butte High (V2016013) Orellan

Cusp terminology for *Eumys cricetodontoides* follows Wood and Wilson

(1936, fig. 2A). Dimensions of mammalian teeth were recorded in mm using a Carrera Precision digital caliper and were photographed using a high-resolution Canon camera system.

Abbreviations. AMNH, American Museum of Natural History, New York, New York; CM, Carnegie Museum of Natural History, Pittsburgh, Pennsylvania; CMN, Canadian Museum of Nature, Ottawa, Ontario, Canada; MCZ, Museum of Comparative Zoology, Harvard University, Cambridge, Massachusetts; PSMP,

Pipestone Springs Main Pocket locality, Montana; **RAM**, Raymond M. Alf Museum of Paleontology, Claremont, California; **YPM-VPPU**, Princeton University collections now housed at the Yale Peabody Museum of Natural History, New Haven, Connecticut; **UNSM**, University of Nebraska State Museum, Lincoln, Nebraska; **USNM**, Department of Paleobiology, United States National Museum of Natural History, Washington D.C.

SYSTEMATIC PALEONTOLOGY

Mammalia Linnaeus 1758

Order Rodentia Brandt 1855

Family Ischyromyidae Alston 1876

Ischyromys Leidy 1856

Ischyromys sp. cf. *I. veterior* Matthew 1903

Figure 3A

Holotype—AMNH 9658 dentary with p4-m3 from Pipestone Springs, Montana (Matthew 1903).

Referred Specimen—RAM 31780 right dentary fragment with m1-2 from RAM locality V2018036.

Description-Discussion—The m1-m2 of RAM 31780 have significant wear so dental features are limited to the general outline of the crown as seen in occlusal view (Figure 3A). When compared directly to dentaries with m1-2 of *Ischyromys veterior* in the RAM collections from Pipestone Springs Main Pocket (PSMP, type locality of *I. veterior*), the m1-2 of RAM 31780 are identical in occlusal outline to those with the same stage of wear. The size in mm of the m1-2 of RAM 31780 (m1 length 3.2, width 3.2, m2 length 3.2, width 3.3) are also similar to RAM specimens of *I. veterior* from PSMP; N=109, m1 length 2.5-3.6, width 2.1-3.6, m2 length 2.6-3.4, width 2.5-3.5 (Homidan and Lofgren 2016). Thus, RAM 31780 is referred to *Ischyromys* sp. cf. *I. veterior*.

Family Cylindrodontidae Miller and Gidley
1918

Cylindrodon Douglass 1901

Cylindrodon sp. cf. *C. fontis* Douglass 1901

Figure 3B-C, Table 1

Lectotype—**CM 738a** partial left dentary with m1-3, Pipestone Springs, Jefferson County, Montana (Douglass 1901).

Referred Specimen—RAM 31779 maxilla fragments with right P4-M3 and isolated left P4 from RAM locality V2018036.

Description-Discussion—These hypsodont teeth are heavily worn (similar to stage of wear of *Cylindrodon fontis* illustrated by Black 1965, in figure 3B) and were recovered from a single concretion. The P4 is square in occlusal outline and has a small anterior and posterior basin and a larger transversely elongate central basin (Figure 3B). The M1-2 are slightly wider than long and have small transversely elongate central basins (Figure 3B-C). The M3 is the smallest tooth and its anterior and central basins are fairly large at this stage of wear. The teeth of RAM 31779 are more hypsodont than *Pseudocylindrodon* and are similar in size to *Cylindrodon fontis* (Table 1). The wear pattern of the P4 and molars of RAM 31779 match that of similarly worn specimens of *Cylindrodon fontis* in the large RAM collections from PSMP (type locality). Emry and Korth (1996, 403) note that “*Cylindrodon fontis* is distinguished from all other species of the genus by the lack of a short, antero-posteriorly oriented loph on the upper molars that divides the basin between the metaloph and posteroloph.” This feature is evident in *C. natronensis* and *C. solarborus* (Emry and Korth 1996, fig. 2A, fig. 4A-C) but is absent from a lightly worn upper dentition of *C. fontis* illustrated by Black (1965, fig. 3D). Wear has obscured whether this loph was present or absent in the molars of RAM 31779 (Figure 3B-C), so the

referral of RAM 31779 to *Cylindrodon* sp. cf. *C. fontis* is tentative.

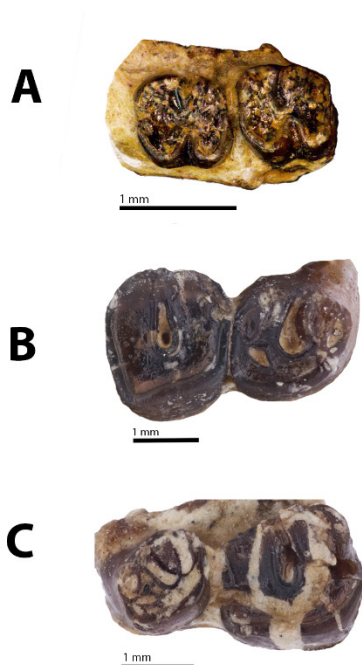


FIGURE 3: *Ischyromys* sp. cf. *I. veterior* and *Cylindrodon* sp. cf. *C. fontis* from Black Butte Middle I. **A.** Occlusal view of RAM 31780 right dentary fragment with m1-2 of *Ischyromys* sp. cf. *I. veterior* from RAM locality V2018036. **B.** Occlusal view of right P4-M1 of RAM 31779 of *Cylindrodon* sp. cf. *C. fontis* from RAM locality V2018036. **C.** Occlusal view of right M2-3 of RAM 31779.

Family Cricetidae Fischer von Waldheim
1817
Eumys Leidy 1856
Eumys cricetodontoides White 1954
Figure 4, Table 2

Holotype—USNM 18748, right dentary with incisor and m1-m3, Canyon Ferry, Lewis and Clark County, Montana (White 1954, fig. 45).

Referred Specimen—RAM 28172 right dentary fragment with damaged m1-3 from RAM locality V2016013.

Description-Discussion—The molars of RAM 28172 are significantly weathered but

TABLE 1. Measurements in mm of the P4-M3 of RAM

31779 compared to the P4-M3 of *Cylindrodon fontis* from PSMP provided by Black (1965, 18). * M2 size increased by calcite infilled cracks.

Tooth		RAM	<i>Cylindrodon.</i>
Site		31779	<i>fontis</i>
P4	L	1.9	1.6-2.0
	W	1.9	1.6-2.7
M1	L	2.0	1.5-2.1
	W	2.1	1.5-2.7
M2	L	2.4*	1.5-2.0
	W	2.6*	1.3-2.4
M3	L	1.7	1.4-1.7
	W	1.5	1.2-1.7

the basic cusp pattern is mostly preserved for m1, less so for m2-3 (Figure 4). *Eumys cricetodontoides* was erected based on the holotype and six other dentaries from Orellan strata near Canyon Ferry, Montana (White 1954) and RAM 28172 is closely comparable to the holotype (USNM 18748) in both morphology and size (Table 2). The elongate m1 of USNM 18748 has a distinct ectolophid with small labial and lingual mesolophids, a posterior cingulum not connected to the entoconid, and a distinct anteroconid and a small protoconid that is united anteriorly with the anteroconid and posteriorly with a large metaconid (White 1954). The m1 of RAM 28172 has these features but its anteroconid is better characterized as a ridge with a slight bulge that denotes the cusp (Figure 4), a condition similar to anteroconid of the holotype as illustrated by White (1954, fig. 45). The longer than wide m2 of the holotype has an anterior cingulum that spans its anterior border, a distinct ectolophid, and a posterior cingulum that is not united with the entoconid (White 1954). The m2 of RAM 28172 also has these features but its ectolophid is faint because the tooth is significantly weathered (Figure 4).

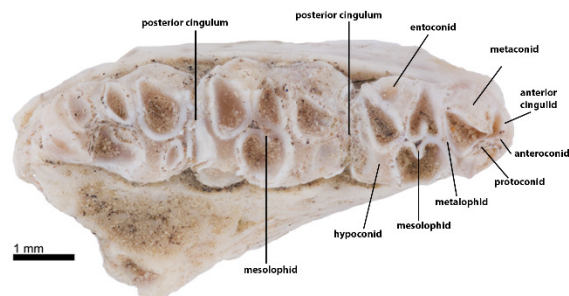


FIGURE 4: *Eumys cricetodontoides* from Black Butte High. Occlusal view of RAM 28172 right dentary with m1-3 of *Eumys cricetodontoides* from RAM locality V2016013.

The Carnegie Museum also made a significant collection of specimens of *Eumys cricetodontoides* in southwest Montana (mostly from Canyon Ferry) about the time White (1954) made his collection for the USNM. These specimens were described by Korth (2011) who emended the diagnosis of *E. cricetodontoides* by including these features; m1 longest lower molar with single connection of metaconid to anteroconid, anterior cingulid strongly developed in m1 anterior to metaconid, and length of m2 approximately equal to m3. The m1-3 of RAM 28172 have these features (Figure 4) and their dimensions are also similar to specimens of *E. cricetodontoides* described by Korth (2011) (Table 2). There is little doubt that RAM 28172 represents *E. cricetodontoides*.

Order Artiodactyla Owen 1848
Family Hypertragulidae Cope 1879
Hypisodus Cope 1873
Hypisodus minimus Cope 1873
Figure 5A-C, Table 3

Holotype—AMNH 6543, maxilla fragments with M1-3, Cedar Creek Member, White River Formation, Colorado (Cope 1873).

Referred Specimens—RAM 28514 left dentary with m2-3 and partial m1 (missing

TABLE 2. Measurements in mm of the m1-3 of RAM 28172 compared to the m1-3 of the holotype of *Eumys cricetodontoides* (USNM 18748) from White (1954, 412) and *Eumys cricetodontoides* from various Orellan localities in Montana provided by Korth (2011, table 3).

Tooth Site		RAM 28172	USNM 18748	Orellan Montana
m1	L	2.7	2.5	2.4-2.9
	W	1.9	1.9	1.8-2.1
m2	L	2.5	2.8	2.2-2.6
	W	2.0	2.2	1.9-2.4
m3	L	2.5	2.5	2.2-2.7
	W	2.0	2.1	1.9-2.3

anterior half of trigonid) from RAM locality V2016010; CM 78277 left maxilla fragment with P4-M2, CM 78278 left dentary with p4-m3, and CM 78279 right dentary with p3-m3 all from CM locality 3527 (10N#2).

Description-Discussion—RAM 28514 represents *Hypisodus*, a relatively rare and uniquely small artiodactyl with hypsodont molars (Meehan and Martin 2004), which in RAM 28514 have been reduced to simple prisms in occlusal view because of wear (Figure 5A). The genotypic species, *Hypisodus minimus*, was erected by Cope (1873) and three additional species were proposed (*H. alacer* Troxell 1920, *H. paululus* and *H. ironis* Cook 1934) but were synonymized with the genotypic species (Webb 1998; Vislobokova 2001; Metais and Vislobokova 2007). More recently, Meehan and Martin (2004) emended the description of *Hypisodus minimus* and erected *H. retallacki*, a new species distinguished by its longer front limbs and larger body size and teeth. The molars of RAM 28514 are similar in size to those of *H. minimus* (Table 3) and RAM 28514 is referred to that species.

RAM 28514 represents the second record of *Hypisodus* from Tertiary strata in southwest Montana. Previously, three well

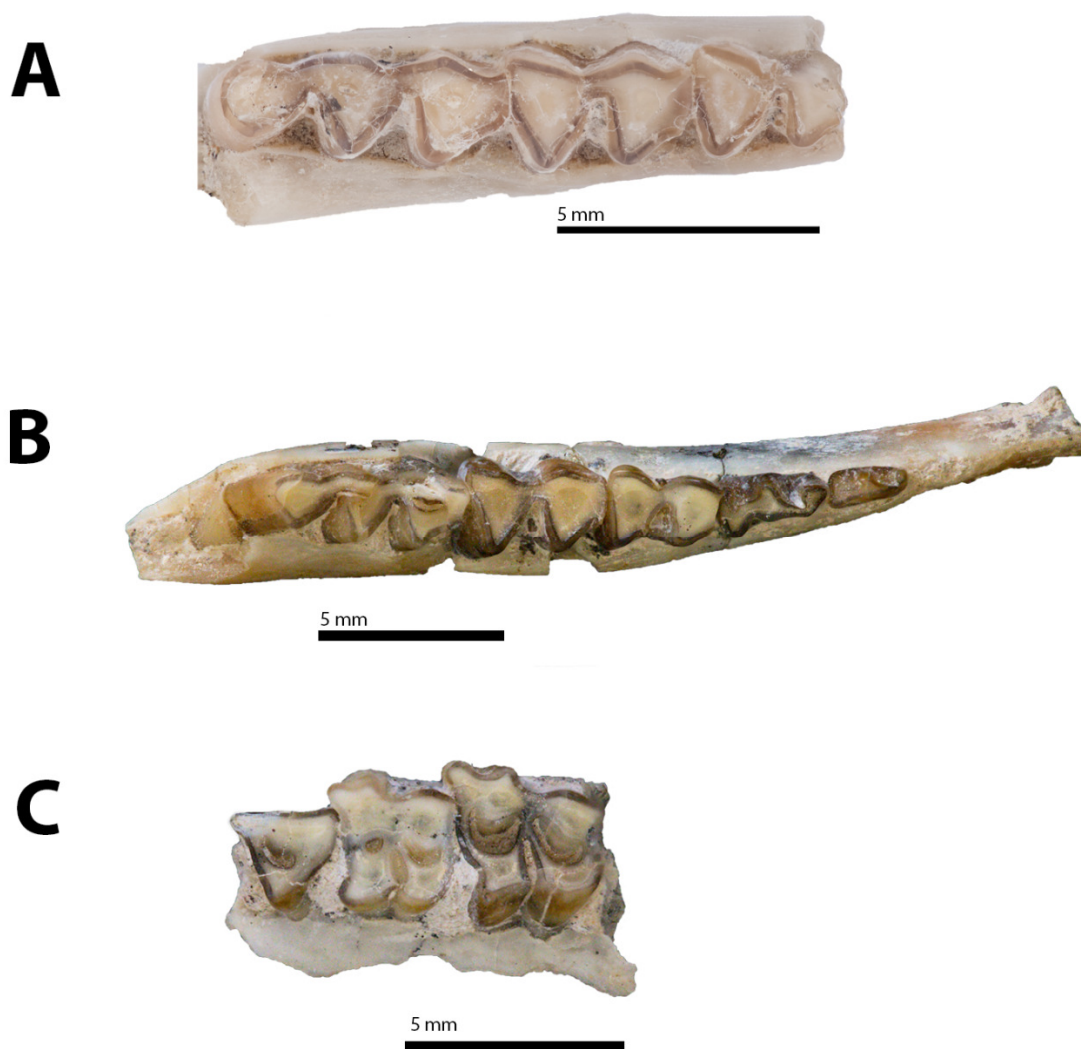


FIGURE 5: *Hypisodus minimus* from Black Butte Middle II and the 10N#2 locality from the Three Forks area. **A.** Occlusal view of RAM 28514 right dentary fragment with m1-3 of *Hypisodus minimus* from RAM locality V2016010. **B.** Occlusal view of CM 78279 right dentary with p3-m3 of *Hypisodus minimus* from CM locality 3527. **C.** Occlusal view of CM 78277 left maxilla fragment with P4-M2 of *Hypisodus minimus* from CM locality 3527.

preserved specimens (CM 78277, CM 78278, CM 78279) of *Hypisodus* were recovered from the late Chadronian 10N#2 site in the Three Forks area (Tabrum 1998; Tabrum et al. 2001). These specimens are also more similar in size to *H. minimus* than *H. retallacki* (Table 3) and CM 78279 (Figure

5B) and CM 78278 exhibit the hypsodont and prismatic crown morphology of the lower dentition of *H. minimus*. CM 78277 is the only upper dentition of *Hypisodus* known from Montana and its molars are hypsodont and have large parastyles that project anterolabially (Figure 5C).

Family Leptomerycidae Zittel 1893

Leptomeryx Leidy 1853

Leptomeryx speciosus Lambe 1908

Figures 6A-F and 7A, Table 4

Holotype—CMN 6311, upper right molar from Bone Coulee, Cypress Hills, Saskatchewan, Canada (Lambe 1908, pl. VIII, fig. 10-11).

Referred Specimens—RAM 31670 left dentary fragment with p4-m3 from RAM locality V2016008; RAM 31743 right maxilla fragment with M1-M2 and anterior margin of M3 from RAM locality V2018036; RAM 28509 right dentary fragment with p4 from RAM locality V2016010; RAM 23870 right dentary fragment with m3 from RAM locality V2017044; YPM-VPPU 30630 right dentary with canine and p3-m3 from Black Butte (precise locality unknown).

Description-Discussion—The taxonomic literature concerning Chadronian-Orellan species of *Leptomeryx* is lengthy (see Shreero et al. 2023 and Welch 2025 for recent reviews). The first comprehensive study was completed by Heaton and Emry (1996), who noted that the morphology of the m3 entoconulid helped delineate two lineages; *L. yoderi*–*L. mammifer* lineage where the m3 entoconulid is a narrow and antero-posteriorly elongate ridge, and *L. speciosus*–*L. evansi* lineage where the m3 entoconulid is an elevated conical cusp (Heaton and Emry 1996, fig. 9; Korth and Diamond 2002). Also, the Orellan species *L. evansi* was differentiated from *L. speciosus* by its highly crenulated molars which also exhibit a *Paleomeryx* fold (Heaton and Emry 1996, fig. 10; Korth and Diamond 2002).

Two of the RAM Black Butte specimens of *Leptomeryx* (RAM 31670, RAM 23870) are dentaries with m3s that have conical and elevated entoconulids (Figure 6A-D) and align with the *L. speciosus*–*L. evansi* lineage. Also, the teeth of both dentaries have slightly crenulated enamel and the molars lack any

trace of a *Paleomeryx* fold which indicates they do not represent *L. evansi*. In large Chadronian samples of *Leptomeryx* where both *L. mammifer* and *L. speciosus* occur, like PSMP in southwest Montana, there is a bimodality in size where the lower cheek teeth of *L. mammifer* are longer than those of *L. speciosus* (Heaton and Emry, 1996, tables 3-5) and the lower cheek teeth of RAM 31670 and RAM 23870 are similar in size to those of *L. speciosus* (Table 4). RAM 28509 is a dentary with p4 and this premolar lacks highly crenulated enamel (Figure 7E-F) and is similar in size to *L. speciosus* (Table 4). The single upper dentition of *Leptomeryx* (RAM 31743; Figure 7G) from Black Butte was found at the same stratigraphic level as RAM 31670 and RAM 23870 and is similar in size to upper molars of *L. speciosus* based on comparison to specimens of *L. speciosus* from the RAM collections. The molars of the dentary of *Leptomeryx* (YPM-VPPU 30630) whose exact Black Butte locality is unknown, have slightly crenulated enamel, lack a *Paleomeryx* fold, and the m3 has a conical and elevated entoconulid (Figure 7A). Also, the p3-m3 of YPM-VPPU 30630 are similar in size to *L. speciosus* from PSMP (Table 4). Thus, the four RAM specimens of *Leptomeryx* and YPM-VPPU 30630 are referred to *L. speciosus*.

Order Perissodactyla Owen 1848

Family Equidae Gray 1821

Mesohippus or *Miohippus* sp.

Figure 7B-D

Referred Specimens—RAM 31742 juvenile left dentary with dp2-3 (missing anterior part of dp2), and RAM 31744 left dentary with p2-p4 (missing posterior part of p4) and crypt for m1 both from RAM locality V2018036.

Description-Discussion—These two equid dentaries are from the same site and one is a juvenile with dp2-3 (RAM 31742)

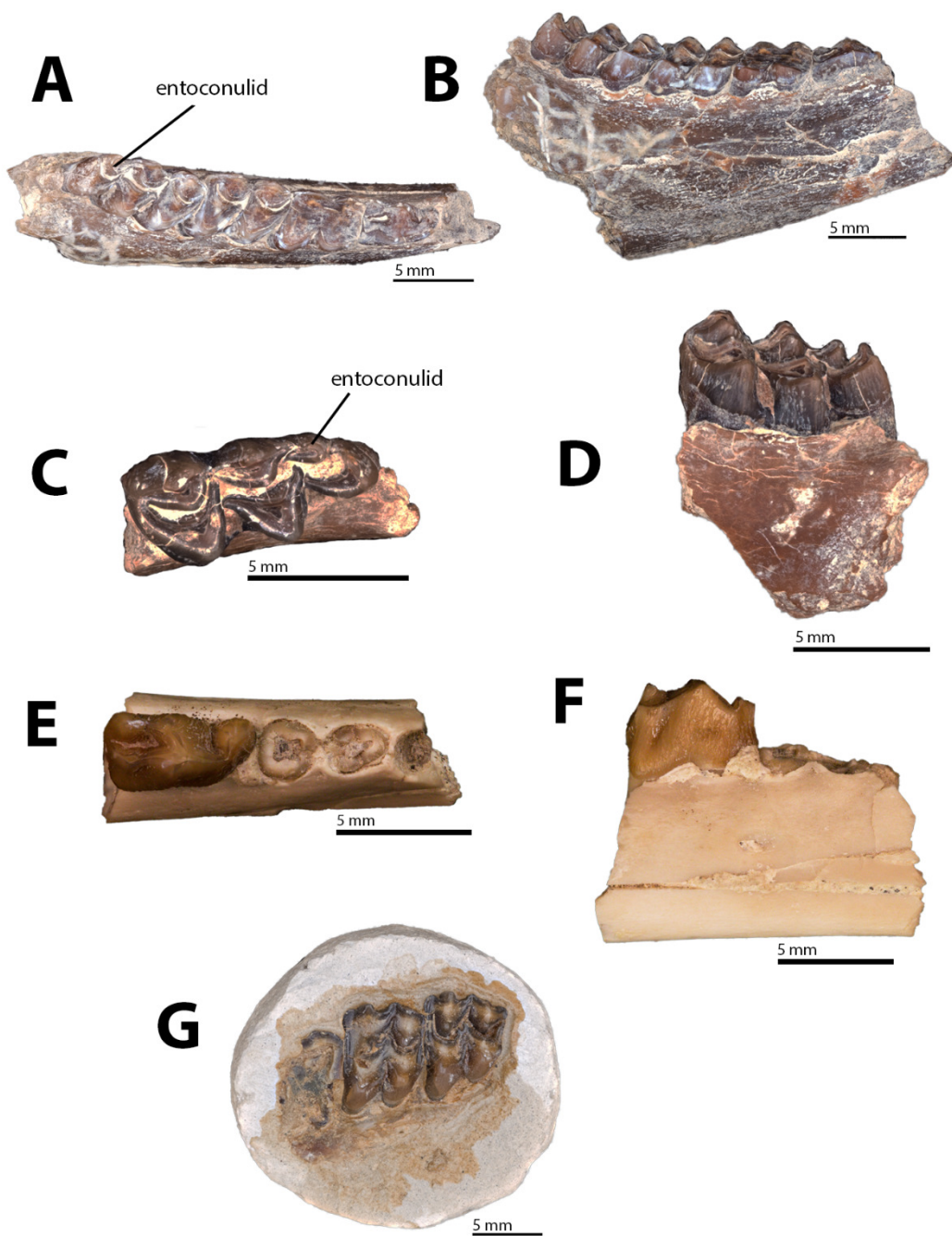


FIGURE 6: *Leptomeryx speciosus* from Black Butte Middle I and Black Butte Middle II. A. Occlusal view of RAM 31670 left dentary fragment with p4-m3 of *Leptomeryx speciosus* from RAM locality V2016008. B. Labial view of RAM 31670. C. Occlusal view of RAM 23870 right dentary fragment with m3 of *Leptomeryx speciosus* from RAM locality V2017044. D. Labial view of RAM 23870. E. Occlusal view of RAM 28509 right dentary fragment with p4 of *Leptomeryx speciosus* from RAM locality V2016010. F. Labial view of RAM 28509. G. Occlusal view of RAM 31743 right maxilla fragment with M1-2 of *Leptomeryx speciosus* from RAM locality V2018036.

and the other (RAM 31744) has p2-4 and what appears to be the crypt for m1 (Figure B-D). Comparison of tooth dimensions of these specimens indicates that deciduous premolars are narrower than permanent premolars but are similar in length; RAM 31744 p2 length 13.8, width 6.9, p3 length 11.1, width 7.8, p4 length 12.4, width 8.8. RAM 31742 dp2 length?, width 5.5, dp3 length 11.2, width 5.9.

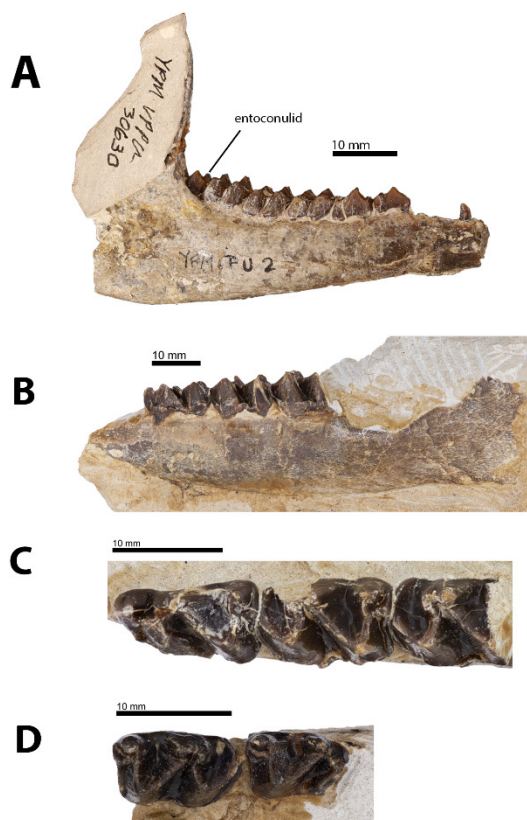


FIGURE 7. *Leptomeryx speciosus* from Black Butte and *Mesohippus* or *Miohippus* sp. from Black Butte Middle I. A. Labial view of YPM-VPPU 30630 right dentary fragment with p3-m3 of *Leptomeryx speciosus* from Black Butte (exact site unknown). B. Labial view of RAM 31744 right dentary fragment with p3-4 of *Mesohippus* or *Miohippus* sp. from RAM locality V2018036. C. Occlusal view of RAM 31744. D. Occlusal view of RAM 31742 left dentary fragment with dp2-3 of *Mesohippus* or *Miohippus* sp. from RAM locality V2018036.

In a recent systematic revision of Oligocene equids (including some late Eocene taxa), species of *Miohippus* and *Mesohippus* were distinguished primarily on features of the upper dentition and *Mesohippus* and *Miohippus* overlap in size (Prothero and Shubin 1989, table 10.3)., so the two fragmentary RAM dentaries could not be confidently referred to either genus.

MAMMALIAN BIOSTRATIGRAPHY AND BIOCHRONOLOGY

Black Butte Middle I and Black Butte Middle II are recently recognized mammalian assemblages at Black Butte that are higher in the section than the Black Butte Low assemblage and lower in the section than the Black Butte High assemblage (Figure 2). The Black Butte Low mammalian assemblage was previously considered late Uintan or early Duchesnean (Lofgren et al. 2020) but five of the seven known taxa are now thought to be shared with the late Uintan (Ui3 interval zone) Dell beds mammalian assemblage from the Sage Creek area of southwest Montana (Lofgren et al. 2025). This suggests that the Black Butte Low mammalian assemblage is late Uintan as well.

Black Butte High is about 100 m above Black Butte Low (Figure 2) and previously yielded three mammalian taxa (*Hesperocyon gregarius*, *Domnina thompsoni*, *Miohippus* sp.), a sparse assemblage that was tentatively considered Chadronian in age (Lofgren et al. 2020). A recently recovered dentary from the site referred to *Eumys cricetodontoides* (RAM 28172) indicates an Orellan age for Black Butte High because the known record of *E. cricetodontoides* is restricted to Orellan rocks in Montana (Korth 2011).

Black Butte Middle I yielded four mammalian taxa (*Ischyromys* sp.cf. *I.*

TABLE 3. Measurements of the p3-m3 and P4-M2 of *Hypisodus minimus* and the holotype of *H. retallacki* (UNSM 1054) from Meehan and Martin (2004; means rounded to tenths) compared to specimens of *H. minimus* from the Gravelly Range (RAM 8514) and CM locality 3527 (10N #2) from the Three Forks area, southwest Montana.

Tooth Site		<i>Hypisodus minimus</i>	<i>Hypisodus retallacki</i>	RAM 28514	CM 78279	CM 78278	CM 78277
p3	L	2.3	2.6	—	2.2	—	—
	W	1.1	1.0	—	1.0	—	—
p4	L	3.0	3.6	—	2.7	3.1	—
	W	1.5	1.7	—	1.2	1.4	—
m1	L	3.3	3.5	2.8	3.0	3.4	—
	W	2.0	2.6	2.3	2.1	2.2	—
m2	L	3.7	4.4	3.3	3.7	3.6	—
	W	2.0	3.1	2.4	2.4	2.3	—
m3	L	5.4	7.3	5.7	5.1	5.6	—
	W	2.0	3.0	2.3	2.0	2.0	—
P4	L	2.7	3.1	—	—	—	2.7
	W	2.0	3.0	—	—	—	2.8
M1	L	3.6	3.6	—	—	—	3.3
	W	2.9	4.0	—	—	—	3.5
M2	L	4.1	4.6	—	—	—	4.2
	W	3.0	4.7	—	—	—	3.7

veterior, *Cylindrodon* sp. cf. *C. fontis*, *Leptomeryx speciosus*, *Mesohippus* or *Miohippus* sp.) from three localities (Figure 2). *Ischyromys veterior* is well known from various sites in Montana which are middle to late Chadronian (PSMP, Fence Pocket, Little Pipestone, West Easter Lily) (Tabrum et al. 1996, 2001; Prothero and Emry 2004). Emry and Korth (1996) noted that *Cylindrodon fontis* appears to be restricted to the “Pipestone Springs fauna” (apparently

referring to specimens from PSMP, Fence Pocket, and Little Pipestone) but may also occur in the McCarty Mountain mammalian assemblage which is early Chadronian (Tabrum et al. 1996, 2001; Prothero and Emry 2004; Hanneman et al. 2025). *Leptomeryx speciosus* first appears in the middle Chadronian and is known from middle and late Chadronian sites in Montana (Tabrum et al. 1996, 2001; Prothero and Emry 2004). The middle Chadronian PSMP

TABLE 4. Measurements in mm of the M1-M2 and p3-m3 of *Leptomeryx speciosus* from the Black Butte Middle sites in the Gravelly Range compared the length of p3-m3 of *L. speciosus* from Pipestone Springs Main Pocket, Montana, provided by Heaton and Emry (1996, table 4).

Tooth Site		<i>Leptomeryx speciosus</i>	RAM 31743	RAM 28509	RAM 23870	RAM 31670	YPM-VPPU 30630
p3	L	5.9-6.6	—	—	—	—	6.3
	W	—	—	—	—	—	2.1
p4	L	6.0-7.0	—	6.6	—	6.1	6.4
	W	—	—	3.3	—	3.5	4.0
m1	L	5.7-7.4	—	—	—	5.3	7.0
	W	—	—	—	—	4.8	4.7
m2	L	6.3-7.8	—	—	—	6.8	7.2
	W	—	—	—	—	5.2	5.3
m3	L	9.5-12.0	—	—	10.0	10.1	10.7
	W	—	—	—	4.2	4.9	4.8
M1	L	—	7.5	—	—	—	—
	W	—	8.9	—	—	—	—
M2	L	—	7.9	—	—	—	—
	W	—	9.0	—	—	—	—

fauna includes *Leptomeryx speciosus*, *Ischyromys veterior*, and *Cylindrodon fontis* (Tabrum et al. 1996, 2001; Prothero and Emry 2004; Lofgren et al. 2017; Hanneman et al. 2022) so it is similar to the sparse Black Butte Middle I mammalian assemblage where identifications of *Ischyromys veterior* and *Cylindrodon fontis* are tentative. Thus, Black Butte Middle I appears to be middle Chadronian.

The Black Butte Middle II mammalian fauna consists of two taxa (*Hypisodus minimus*, *Leptomeryx speciosus*) from V2016010 (Figure 2). *Hypisodus minimus* is known from late Duchesnean to early Arikareean strata in Colorado, Nebraska, South Dakota and Montana (Meehan and Martin 2004). The dentary of *Hypisodus minimus* (RAM 28414) from the Gravelly Range is the second known occurrence of

Hypisodus from Tertiary strata in southwest Montana. The other occurrence is the Three Forks area, primarily from the 10N#2 locality of late Chadronian age (Tabrum 1998). Besides *Hypisodus*, Tabrum et al. (2001) noted that the 10N#2 site yielded other mammalian taxa not found from other middle or late Chadronian sites in southwest Montana. For example, the massive mammalian sample from PSMP does not include a single specimen identified as *Hypisodus*. Recovery of *H. minimus* from the Gravelly Range indicates a greater presence of this tiny artiodactyl in southwest Montana than previously thought. *Leptomeryx speciosus* is the other taxon identified from Black Butte Middle II and it has a biochronologic range of middle to late Chadronian (Prothero and Emry 2004). The biochronologic range of *Hypisodus minimus*

is broad, but the presence of *L. speciosus* at Black Butte Middle II indicates that the site is middle-late Chadronian.

The sparse mammalian assemblages above the stratigraphic level of Black Butte Low indicate that these strata are middle Chadronian (Black Butte Middle I), middle-late Chadronian (Black Butte Middle II), and Orellan (Black Butte High). Documentation of a sequence of four superposed levels of mammalian assemblages at Black Butte is fortuitous as Gravelly Range Tertiary strata are sparsely fossiliferous (Lofgren et al. 2020). The current list of taxa known from mammalian assemblages in Gravelly Range strata are in ascending stratigraphic order:

Black Butte Low/Rapamys Site, late Uintan, *Rapamys atramontis*, *Pareumys muffleri*, *Lycophocyon tabrumi*, *Diplobunops matthewi*, *Protoreodon pumilus*, *Triplopus* sp. cf. *T. rhinocerinus*, and *Hesperaletes borineyi*.

Teepee Mountain, late Duchesnean-early Chadronian, *Megalagus* sp. cf. *M. brachyodon*, *Palaeolagus* sp., *Ischyromys douglassi*, *Paradjidaumo disjunctus*, and *Diplobunops matthewi*.

Black Butte Middle I, middle Chadronian, *Ischyromys* sp. cf. *I. veterior*, *Cylindrodon* sp. cf. *C. fontis*, *Leptomeryx speciosus*, and *Meshippus* or *Miohippus*.

Black Butte Middle II, middle-late Chadronian, *Hypisodus minimus* and *Leptomeryx speciosus*.

Black Butte High, Orellan, *Eumys cricetodontoides*, *Hesperocyon gregarius*, *Domnina thompsoni*, and *Miohippus* sp.

Lion Mountain High, Whitneyan, *Herpetotherium fugax*, *Nanodelphys* sp. cf. *N. hunti*, *Palaeolagus burkei*, *Eumys elegans*, *Leidymys*, *Ocetocyon cuspidatus*, *Centetodon magnus*, *Hypertragulus calcaratus*, and *Diceratherium tridactylum*.

The ages of the strata that yielded these mammalian assemblages appear to indicate that there was not a major hiatus in Gravelly

Range sedimentation throughout the middle Eocene to early Oligocene (late Uintan-Whitneyan). Recognition of two new mammalian assemblages from Black Butte strata suggests that procurement of additional specimens from other outcrop areas, especially the thick Lion Mountain section, would result in further refinement in determining the age of Tertiary sedimentation in the Gravelly Range.

SUMMARY

Based on recently recovered specimens from Eocene-Oligocene strata at Black Butte in the Gravelly Range, new records of rodents (*Ischyromys* sp. cf. *I. veterior*, *Cylindrodon* sp. cf. *C. fontis*, *Eumys cricetodontoides*) and artiodactyls (*Hypisodus minimus*, *Leptomeryx speciosus*) are known from two new mammalian assemblages, Black Butte Middle I and Black Butte Middle II. These assemblages occur in strata underlain by the Black Butte Low assemblage now considered late Uintan and overlain by the Black Butte High assemblage now considered Orellan because of the addition of *Eumys cricetodontoides*. Black Butte Middle I strata yielded *Ischyromys* sp. cf. *I. veterior*, *Cylindrodon* sp. cf. *C. fontis*, and *Leptomeryx speciosus*, taxa best known from middle Chadronian rocks in southwest Montana. Black Butte Middle II strata yielded two taxa, *Hypisodus minimus* which has a broad biochronologic range and is only known elsewhere in Montana from late Chadronian rocks in the Three Forks area, and *Leptomeryx speciosus* which is known from various middle-late Chadronian sites in Montana. Based on the presence of *L. speciosus*, Black Butte Middle II strata are considered middle-late Chadronian. The ascending stratigraphic order of mammalian assemblages at Black Butte are, Black Butte Low, Black Butte Middle I, Black Butte Middle II, and Black Butte High (Figure 2). Documentation of late Uintan-Orellan

mammalian assemblages at Black Butte combined with the ages of previously identified assemblages at Lion Mountain, Teepee Mountain, and the *Rapamys* Site indicate that Tertiary sedimentation in the Gravelly Range area apparently occurred throughout the middle Eocene-early Oligocene without a major hiatus.

ACKNOWLEDGMENTS

We thank C. Wideman, D. Mirkin, K. Piacentini, D. Everhart, C. Lachman, M. Lachman, and W. Lachman for aid in the field; the Ennis Ranger District and the Supervisors Office of the Beaverhead-Deerlodge National Forest for collection permits and field support, especially D. Wheeler, D. Olson, G. Ash, and E. Torgerson; M. Stokes, J. Heuck, and A. Colon for preparation of specimens; V. Rhue and D. Brinkman from the YPM and S. Davis and A. Heinrich of the CM for access to collections and/or loan of specimens; and the David B. Jones Foundation, A. Farke and the Raymond M. Alf Museum of Paleontology, and the Mary Stuart Rogers Foundation for financial support.

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